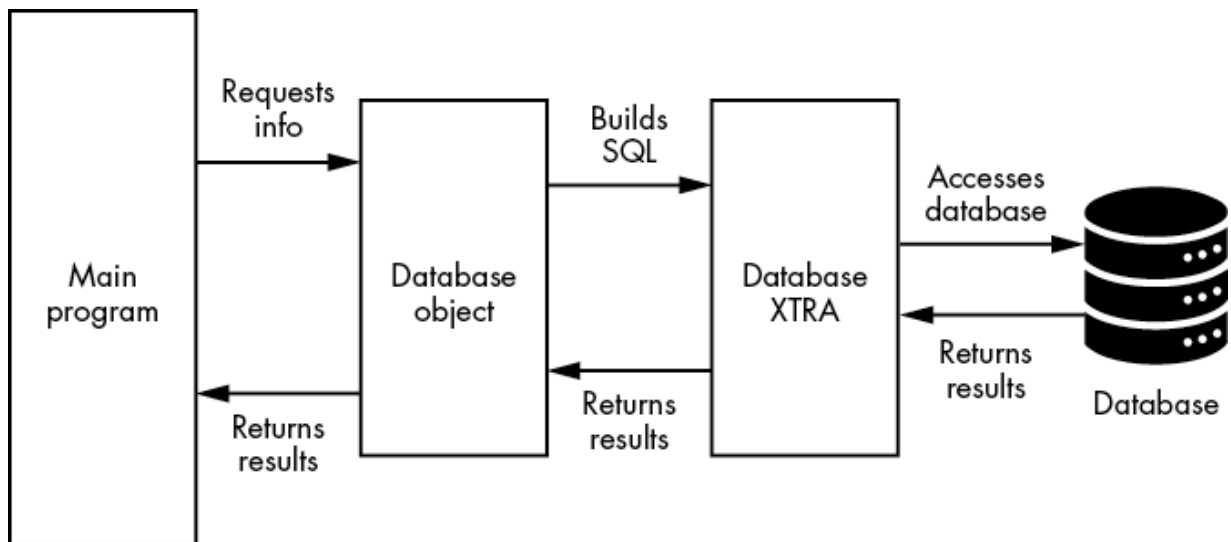


A Story from the Real World

A number of years ago, I was involved in the design and development of a very large educational project that was built in an environment called *Director* from Macromedia (later Adobe), using the object-oriented *Lingo* language. Director was designed to be extended through *XTRAs* that could add functionality, similar to the way plug-ins are added to browsers. These XTRAs were developed and sold by a number of third-party vendors. In the design, we planned on storing navigational and other course-related information in a database. I looked at all the different database XTRAs that were available and purchased a particular XTRA, which I'll call XTRA1.

Each XTRA came with documentation of its API, which showed how to make queries to the database using Structured Query Language (SQL). I decided to create a `Database` class that incorporated all the functionality of accessing the database using XTRA1's API. That way, all code that communicated with the XTRA directly was in the `Database` class. [Figure 8-1](#) shows the overall architecture.



[Figure 8-1](#): The architecture of accessing a database using an object and an XTRA

When the program started, it created a single instance of the `Database` class. The main code was a client of the `Database` object. Whenever the main code wanted information from the database, rather than formatting a SQL query itself, it called a method of the `Database` object, supplying details about what information it wanted. The methods in the `Database` object translated